Joong-Gil Kim

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Research Interests

My research goal is to build robots that learn complex control skills in simulation and carry them over to real hardware without losing reliability. I am particularly interested in reinforcement learning for proprioceptive reactive manipulation with dexterous hands, and in the high-fidelity physics simulation that makes such policies transfer to the real robot. I am also interested in locomotion and humanoid whole-body control, with a longer-term interest in physical intelligence for entire robot systems.

Education

Korea University of Technology and Education Mar 2025 – Feb 2027 (expected)
M.S. in Electronic Engineering (Advisor: Yong-Jae Kim) *Cheonan, Korea*

Korea University of Technology and Education Mar 2020 – Feb 2025
B.S. in Mechatronics Engineering *Cheonan, Korea*

Research Experience

IRIM Lab, Korea University of Technology and Education Sep 2024 – Present
Research Assistant (Advisor: Yong-Jae Kim) *Cheonan, Korea*

- Simulation lead for ALLEX, a humanoid upper-body robot with dexterous hands: built its simulation framework and sim-to-real pipeline, including nonlinear motor–joint modeling.
- Transferred a proprioceptive reactive manipulation RL policy to the real ALLEX.

Neuromeka Jul 2024 – Aug 2024
Research Intern *Pohang, Korea*

- Built smart-factory simulation environments in NVIDIA Isaac Sim.

Control and Dynamic Systems Lab, Korea University of Technology and Education Sep 2023 – Aug 2024
Undergraduate Research Assistant (Advisor: Hong-keun Kim) *Cheonan, Korea*

Honors and Awards

Bronze Award (3rd Place), 2nd Koh Young AI Competition 2024
Held at ICCAS 2024, Jeju, Korea. For the 3-DOF inverted pendulum swing-up and stabilization project.

Grand Prize, UCLA RoMeLa Director's Award, Robot Utilization Idea Competition 2023
1st place out of 99 teams nationwide. For the Reusable Sky-Jump Robot.

3rd Place, Korea Association of Robot Industry Chairman's Award, R-BIZ Challenge 2023
ZEUS Industrial Robot ZERO Mission. For the Toppled Vial Cylinder Removal Robot.

Publications

[4] Taemoon Jeong, Doyoung Kim, Sun Woo Kim, **Joong-Gil Kim**, Seunghean An, Yoonbyung Chai, Chanwoo Kim, Taehyun Byun, Sangseung Kang, Youngwoo Yoon, Yong-Jae Kim, Xue Bin Peng, Kyungjae Lee, and Sungjoon Choi.
FAMoR: Fine-grained Agentic Motion Retargeting via Local Refinement, IEEE International Conference on Robotics and Automation (ICRA), 2027 *(under review)*.

[3] Juhan Park, Yeoncheol Jang, **Joong-Gil Kim**, Junha Cha, Hansol Park, Kyungjae Lee, Yong-Jae Kim, and Sungjoon Choi.
Unifying Fingertip Motion and Joint Posture for Dexterous Hand Retargeting, IEEE International Conference on Robotics and Automation (ICRA), 2027 *(under review)*.

[2] **Joong-Gil Kim**, Wontae Ye, Geunwoo Cho, Seong-Ho Yun, Se-Hyoung Cho, and Yong-Jae Kim.
Comparative Study on Agility, Efficiency, and Impact Absorption of Bipedal Robots with Active Toes, International Conference on Control, Automation and Systems (ICCAS), 2026.

[1] Juhan Park, Taerim Yoon, Seungmin Kim, **Joong-Gil Kim**, Wontae Ye, Jeongeun Park, Yoonbyung Chai, Geonwoo Cho, Geunwoo Cho, Dohyeong Kim, Kyungjae Lee, Yong-Jae Kim, and Sungjoon Choi.
Learning Dexterous Grasping from Sparse Taxonomy Guidance, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

Academic Service

IEEE International Conference on Robotics and Automation (ICRA) 2027 (in progress)
Reviewer

Projects

ALLEX Simulation Model Release Jun 2026
Simulation lead, with WIRobotics

- Public release of the ALLEX simulation model, validated against the physical robot so that simulated and real behavior match down to contact forces.

3-DOF Inverted Pendulum Oct 2024
Team Pluto X, KOREATECH

- Swing-up and stabilization control of a three-degree-of-freedom inverted pendulum.

Reusable Sky-Jump Robot Dec 2023
Team project, KOREATECH

- Proposed a reusable design for the Sky-Jump jumping robot, extending the concept introduced by Prof. Dennis Hong (UCLA RoMeLa).

Toppled Vial Cylinder Removal Robot Oct 2023
Team project, KOREATECH

- Vision-guided robot that detects vials toppled over on a pharmaceutical vial-transfer conveyor line and removes them automatically, keeping the packaging process running without manual intervention. [video]

Technical Skills

Programming	Python (primary); MATLAB, C
Simulation	Isaac Sim, Isaac Lab, MuJoCo, Genesis, Newton
GPU Computing	NVIDIA Warp
Learning	PyTorch, reinforcement learning, imitation learning
Tools	Linux, Git, ROS 2, AWS